

EdinburghAI

Teaching a Computer to Guess

An introduction to Machine Learning, Linear
Regression and Neural Networks

THE CORE LOOP

Same five steps, every time.



01. Guess

Make a prediction



02. Measure

Score how wrong



03. Improve

Nudge it better



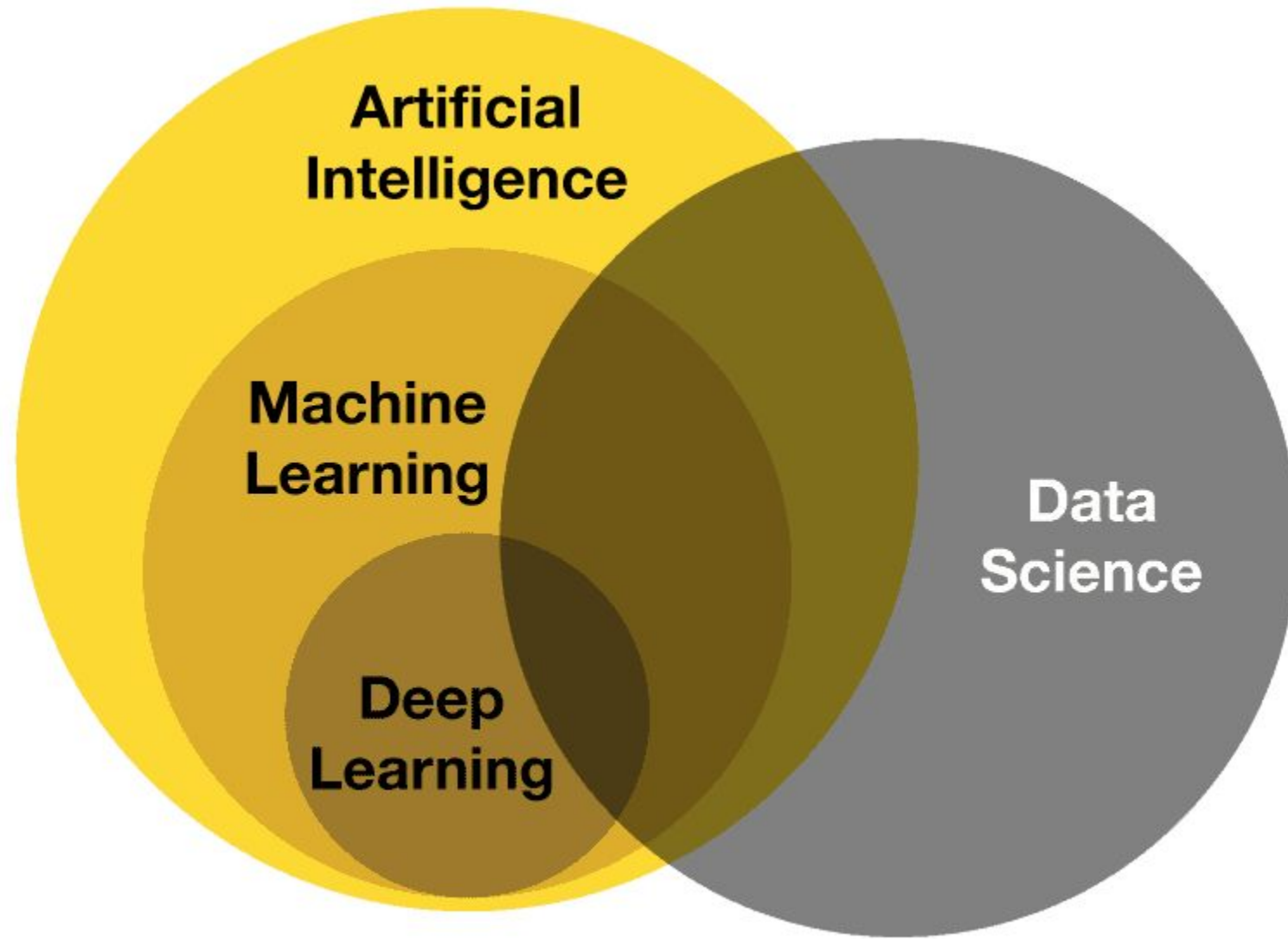
04. Predict

Use the model



05. Scale

...and that is ML



YOU ALREADY USE THIS

Machine Learning in Everyday Life

Every one of these is the same core idea we're about to build from scratch.



LLMs & Chatbots

ChatGPT, Claude, Gemini



Image Recognition

Face ID, searching photos



Recommendations

Netflix, Spotify, TikTok

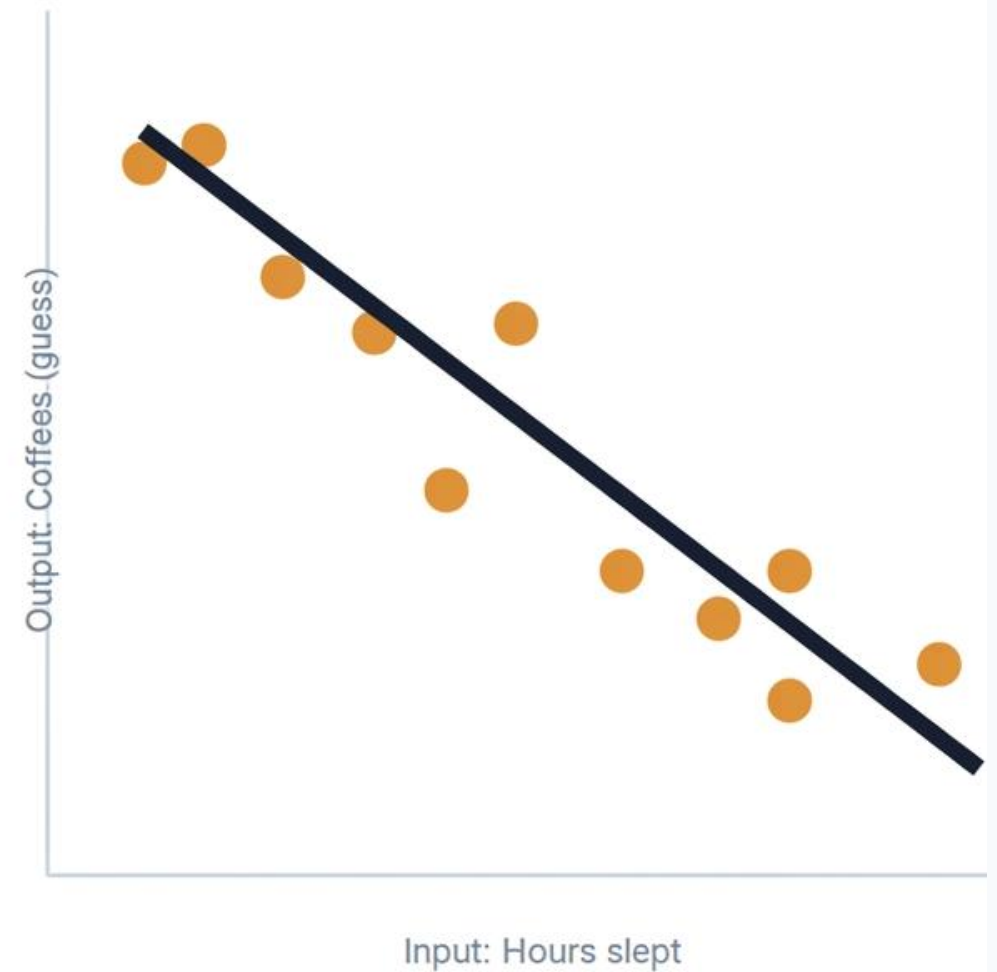
STEP 01 • GUESS

A model is just a rule for guessing

The simplest possible rule is a straight line.

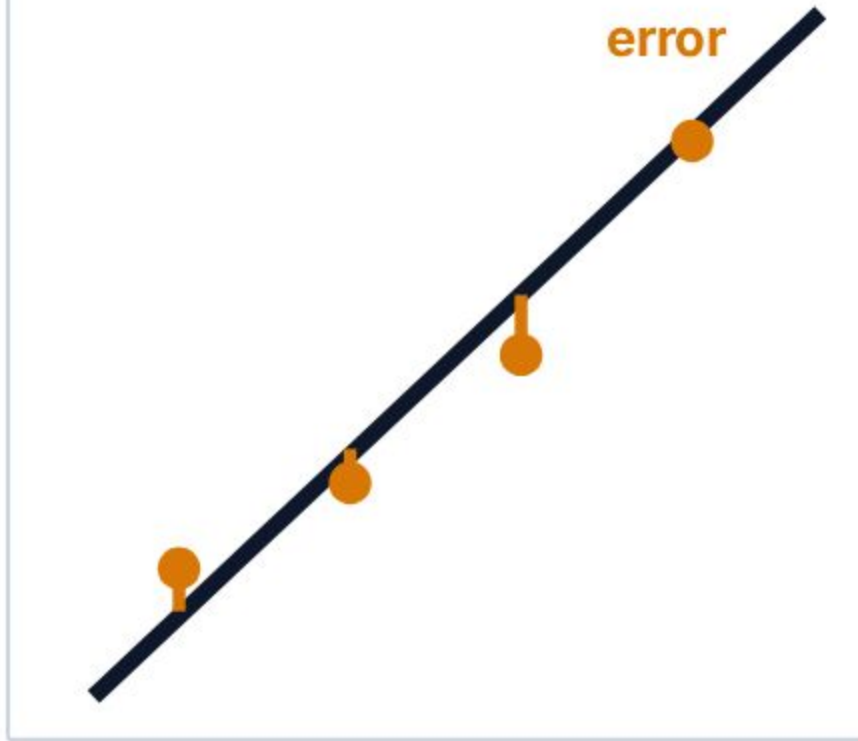
$$y = mx + c$$

- **m = slope:** How steep the line is.
- **c = intercept:** Where the line starts.
- **Training:** Finding the m and c that fit best.



STEP 02 · MEASURE

How wrong is a line? Measure the gaps



$$\text{Error} = (\text{Predicted} - \text{Actual})^2$$

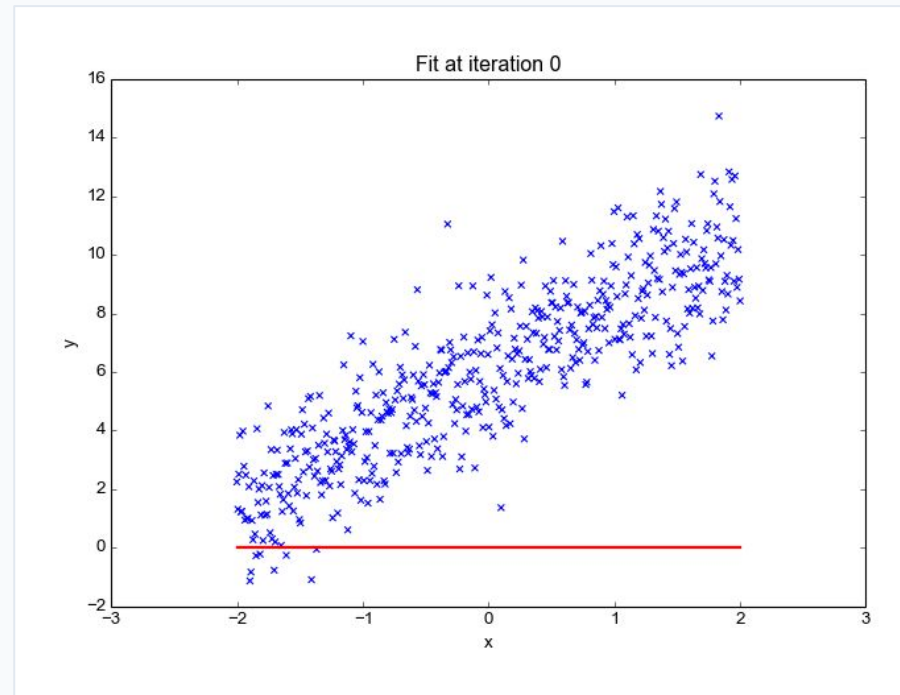
Square the errors: the loss function

- **MSE:** Average of $(\text{prediction} - \text{actual})^2$. One number that says how wrong the whole line is. Lower = better.
- **Squaring kills the sign:** Errors of +2 and -2 no longer cancel to zero. Both count as "wrong."
- **Big misses hurt more:** A gap of 4 costs 16. The line is pushed hard to avoid outliers.

STEP 03 · IMPROVE

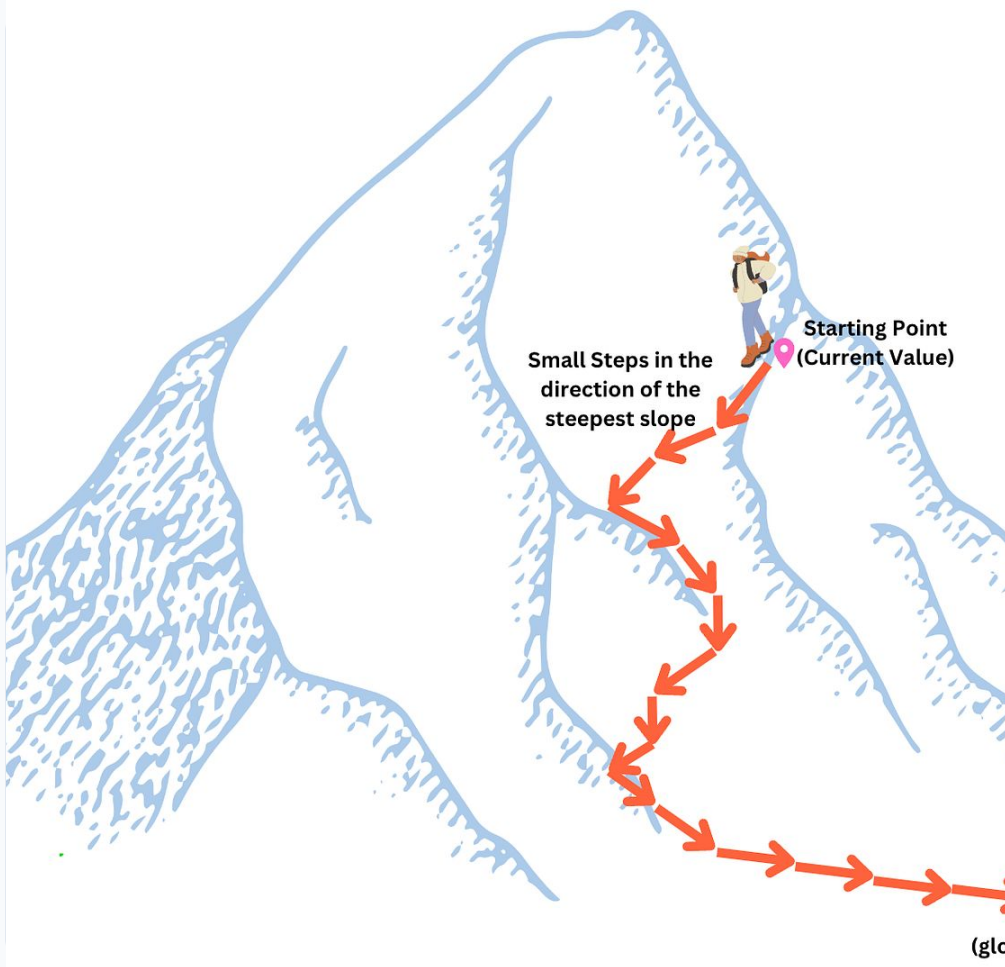
Watching the model learn

Each frame nudges the line a little closer to the data, one gradient descent step at a time.



Gradient descent: walk downhill blindfolded

You're on a foggy hill and can only feel the slope under your feet. To reach the bottom: feel which way is down, take a small step, repeat.



1 Feel the slope

1

The gradient points uphill, so go the other way.

2 Take a small step

2

Step size = the "learning rate".

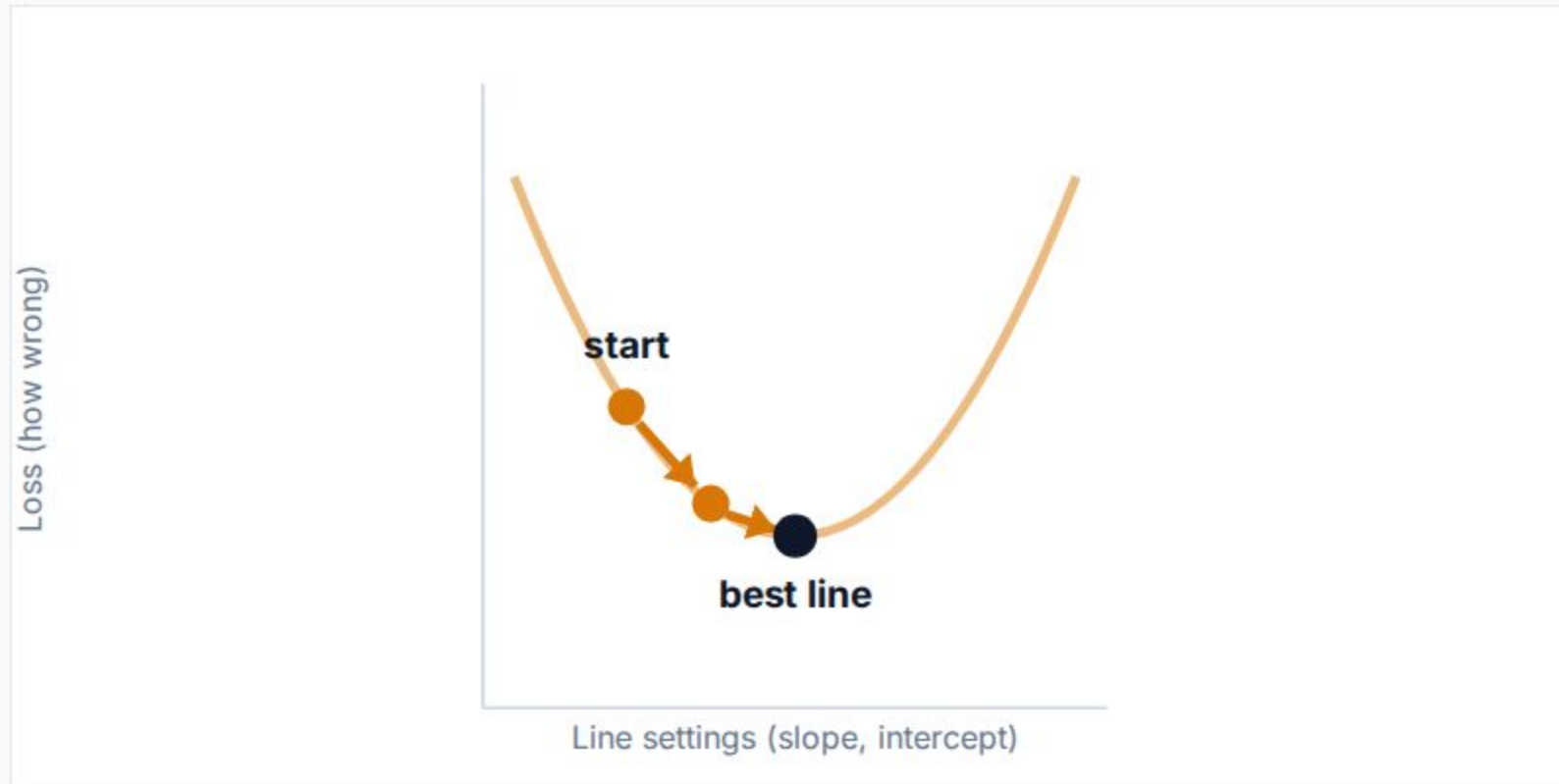
3 Repeat

3

Each step lowers the loss until you settle at the bottom.

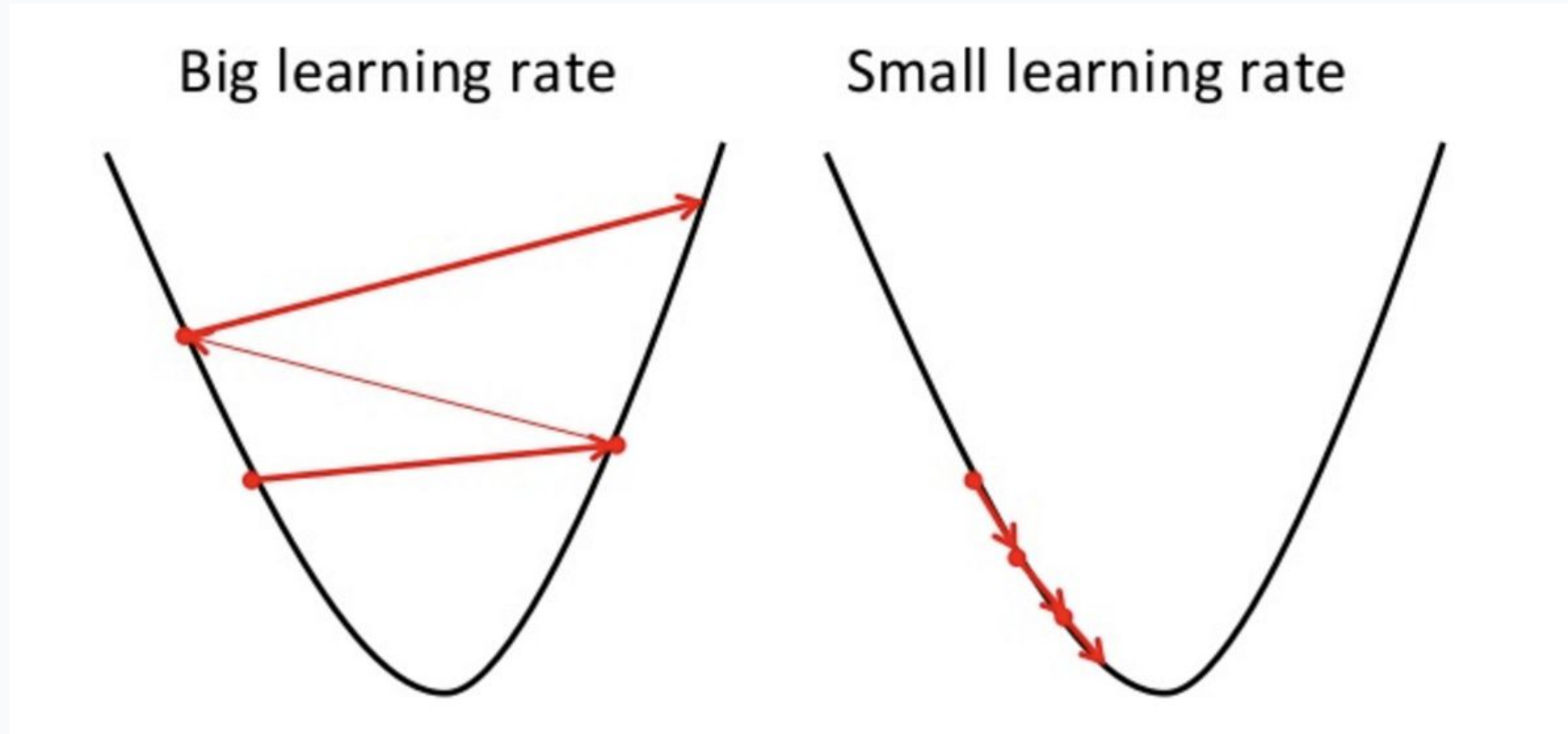
THE LOSS VALLEY

Every possible line is a point on a map



STEP 03 · IMPROVE

Choosing a learning rate



THE LIMITS OF A STRAIGHT LINE

What if the data curves?

Real-world relationships rarely follow a perfect straight line.

- **The Fix:** Add non-linear terms (like x^2 , x^3) directly to the input data.
- **The Magic:** Standard linear regression math, but it draws a curve.
- **The Catch:** Great for simple curves, fails on wildly complex interactions.



$$y = m_1 x + m_2 x^2 + c$$

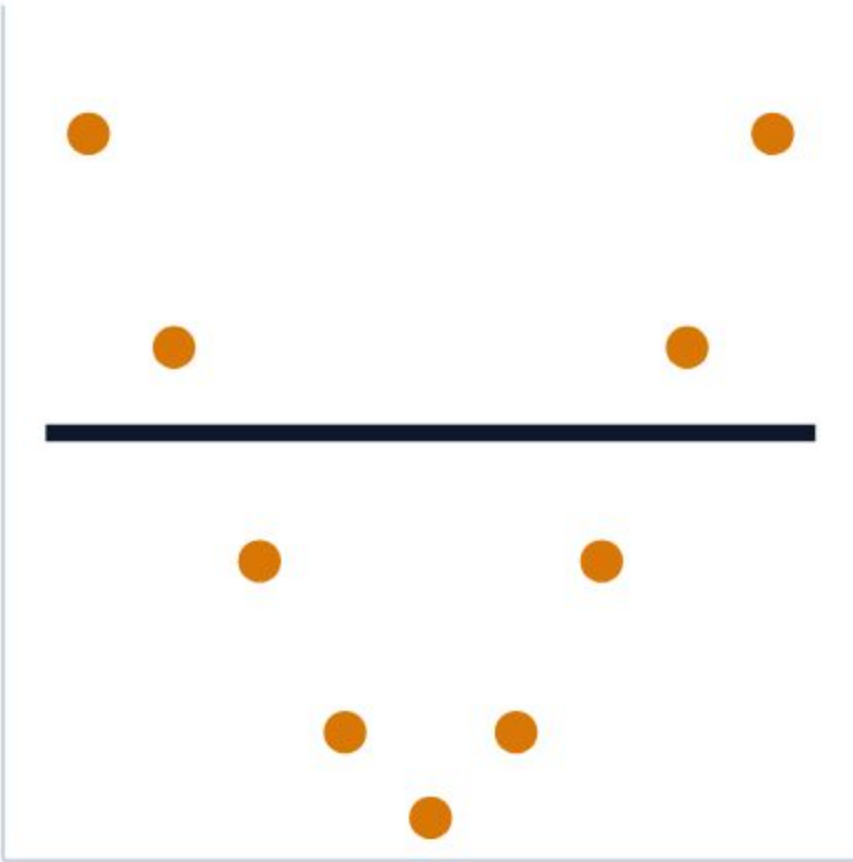
STEP 04 & 05 · PREDICT & SCALE

Same loop, bigger middle box

Linear regression is just a line with 2 knobs. It fails completely when relationships curve or interact unpredictably.

- **Neural Networks:** Millions of knobs arranged in layers, separated by non-linear bends (activation functions).
- **LLMs:** Billions of knobs.

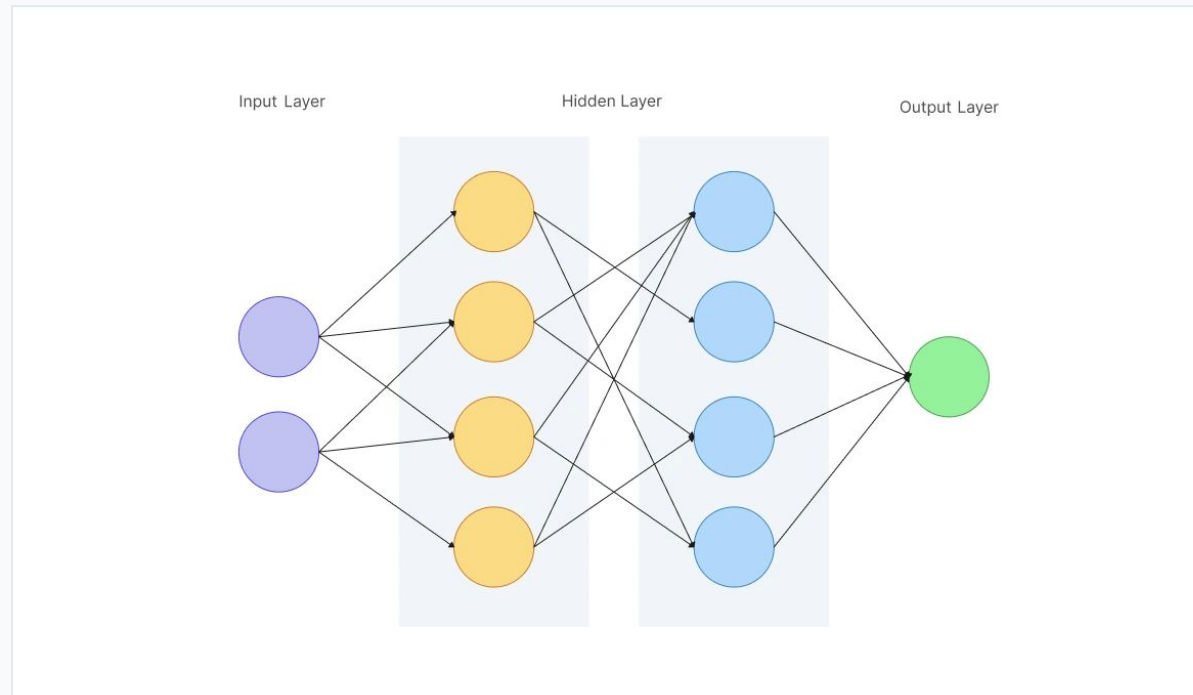
Data → Predict → Measure → Improve → Repeat. That's the whole game.



STEP 04 & 05 · PREDICT & SCALE

What a neural network looks like

Inputs flow forward through hidden layers of weighted connections to produce an output.



A network of straight lines

Neural networks are just thousands of linear regressions stacked together. But stacking them isn't enough.

- **The Collapse Problem:** Stacking linear equations mathematically collapses into one giant straight line.
- **The Fix (Activation):** A mathematical gate (like ReLU) added to every neuron to introduce a non-linear bend.
- **The Result:** Thousands of bent lines combine to learn wildly complex real-world shapes.

Inside One Neuron

1. LINEAR REGRESSION

$$y = \sum wx + b$$

2. ACTIVATION FUNCTION



Get the notebook on Kaggle

1. Import Notebook

1. Go to Kaggle.com (sign in/create account)
2. Top-left: Create → Notebook → File → Import Notebook
3. Choose the GitHub tab
4. Search: EdinburghAI/workshops
5. Pick your level (Beginner/Intermediate)
6. Click Import

✕ Import notebook

File Colab Link **GitHub** Code MCP

Q EdinburghAI/workshops ✕

☐ Include private repositories

BRANCH
main

FILE
File

TheoreticalWorkshops/Sem2Workshop1/yolo-workshop-api/model_backbone.py (3.21 KB)

TheoreticalWorkshops/Sem2Workshop1/yolo-workshop-api/results_viewer.py (1.58 KB)

TheoreticalWorkshops/Sem2Workshop2/LSTM-Solved.ipynb (65.04 KB)

TheoreticalWorkshops/Sem2Workshop2/LSTM.ipynb (67.88 KB)

Workshops2026-27/Sem1Workshop1/IntroToML-Beginner.ipynb (17.13 KB)

Workshops2026-27/Sem1Workshop1/IntroToML-Intermediate.ipynb (17 KB)

Workshops2026-27/Sem1Workshop1/IntroToML-Solved.ipynb (19.06 KB)

Questions? Come say hi.

Thanks for playing along. You came in guessing. You're leaving with the core of machine learning.

hello@edinburghai.org